

The era of hyper-scaling in electronics

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In the past five decades, the semiconductor industry has gone through two distinct eras of scaling: the geometric (or classical) scaling era and the equivalent (or effective) scaling era. As transistor and memory features approach 10 nanometres, it is apparent that room for further scaling in the horizontal direction is running out. In addition, the rise of data abundant computing is exacerbating the interconnect bottleneck that exists in conventional computing architecture between the compute cores and the memory blocks. Here we argue that electronics is poised to enter a new, third era of scaling — hyper-scaling — in which resources are added when needed to meet the demands of data abundant workloads. This era will be driven by advances in beyond-Boltzmann transistors, embedded non-volatile memories, monolithic three-dimensional integration and heterogeneous integration techniques.

The invention and demonstration of the self-aligned planar-gate silicon metal–oxide–semiconductor field-effect transistor (MOSFET) in the late 1960s created a rock-solid foundation for the semiconductor integrated circuit industry. Gordon Moore's bold prediction that transistor count would double every two years, in conjunction with Robert Dennard's scaling guidelines, subsequently led to the exponential growth of the integrated circuit industry, as transistor dimensions shrunk until 2000^{1,2}. This period is often referred to as the era of geometric (or classical) scaling.

After the year 2000, as geometric scaling slowed down, a second era of equivalent (or effective) scaling emerged, aided by the introduction of the strained silicon and silicon–germanium channels, high- κ /metal-gate stack and non-planar fin field-effect transistors (FinFETs). During this period, the effective velocity of electrons and holes in the channel (strain) increased³, the effective oxide thickness (high- κ dielectric) decreased⁴ and the effective width of the minimum-size transistor (fin height, fin pitch) increased⁵ — and the integrated circuit industry continued with its forward exponential march of doubling transistor count every two years. High-aspect-ratio tall fins, coupled with fully self-aligned source–drain contacts and self-aligned gate contact over active diffusion area, further accelerated the rate of increase in transistor density. Innovations in both process integration and advanced patterning techniques increasingly allowed designers to target standard cell designs with low track height⁶. The track height of a standard cell is determined by the horizontal metal line pitch times the number of metal tracks needed for the power and ground rails, for the input and output pins and for intra-cell routing. We expect the era of equivalent (or effective) scaling to continue until 2025, enabled by innovative design-technology co-optimization (DTCO) efforts and process innovations such as buried rails, super vias, self-aligned multi-patterning, the introduction of extreme UV lithography, and eventually high-numerical-aperture extreme UV lithography.

But what happens beyond 2025? The semiconductor electronics community is challenged with this daunting question, as scaling needs to enter an uncharted territory beyond geometric and effective scaling. We believe that this uncertainty provides a unique opportunity to orchestrate a shift in the way electronics is implemented. Specifically, we suggest that electronics will enter a new (third) era — called hyper-scaling — enabled by the functional augmentation of today's technology (Fig. 1). Hyper-scaling refers to the ability of a technology to efficiently scale from a few billion to a trillion

components on chip, depending on the demands of the workload. The era of hyper-scaling will be fuelled by innovations in four major areas: beyond-Boltzmann transistors; embedded, high-performance memories beyond static random-access memory (SRAM) and dynamic random-access memory (DRAM); monolithic 3D integration of logic, memory, analogue and I/O transistors; and heterogeneous integration of functionally diverse integrated circuits delivering monolithic-like performance.

Breaking the Boltzmann barrier

As transistor scaling has progressed, it has become increasingly difficult to improve the intrinsic performance of the basic building block: the transistor. As a result, significant research and development effort has been focused on device circuit co-design to extract the final few drops of efficiency through efficient DTCO. Despite this, one truth remains: if the performance of the transistor can be improved, it can immediately and significantly improve efficiency at every level. As a result, we opine that research on new ideas for transistors will continue.

The performance and energy efficiency of the transistor is determined by the ON-state current, and the ON-state to OFF-state current ratio at a given supply voltage of operation. There exists a minimum-allowable operating supply voltage so as to prevent an unacceptable increase in the OFF-state current (I_{OFF}) while guaranteeing an acceptable ON-state current (I_{ON}). There are two complementary approaches toward scaling the minimum supply voltage of a transistor, while keeping the I_{ON} to I_{OFF} current ratio constant. The first approach involves the incorporation of germanium⁷, group III–V compound semiconductors⁸ and carbon nanotubes⁹ as beyond-silicon channel materials with higher intrinsic carrier mobilities and faster top-of-the-barrier injection velocities. Higher mobilities and velocities allow these transistors with non-silicon channels to operate with high on-state current at low gate overdrive voltage (that is, the amount of gate voltage above the threshold voltage) and enable high-speed operation at lower supply voltages than their silicon counterpart. The second approach is related to improvement of the so-called subthreshold swing of the transistor. The swing is the amount of gate voltage required to change the source to drain current in the channel of a MOSFET by one order of magnitude. It turns out that the Boltzmann distribution of electrons at the source and drain of the MOSFET limits the subthreshold swing to a minimum value of 60 mV per decade. In other

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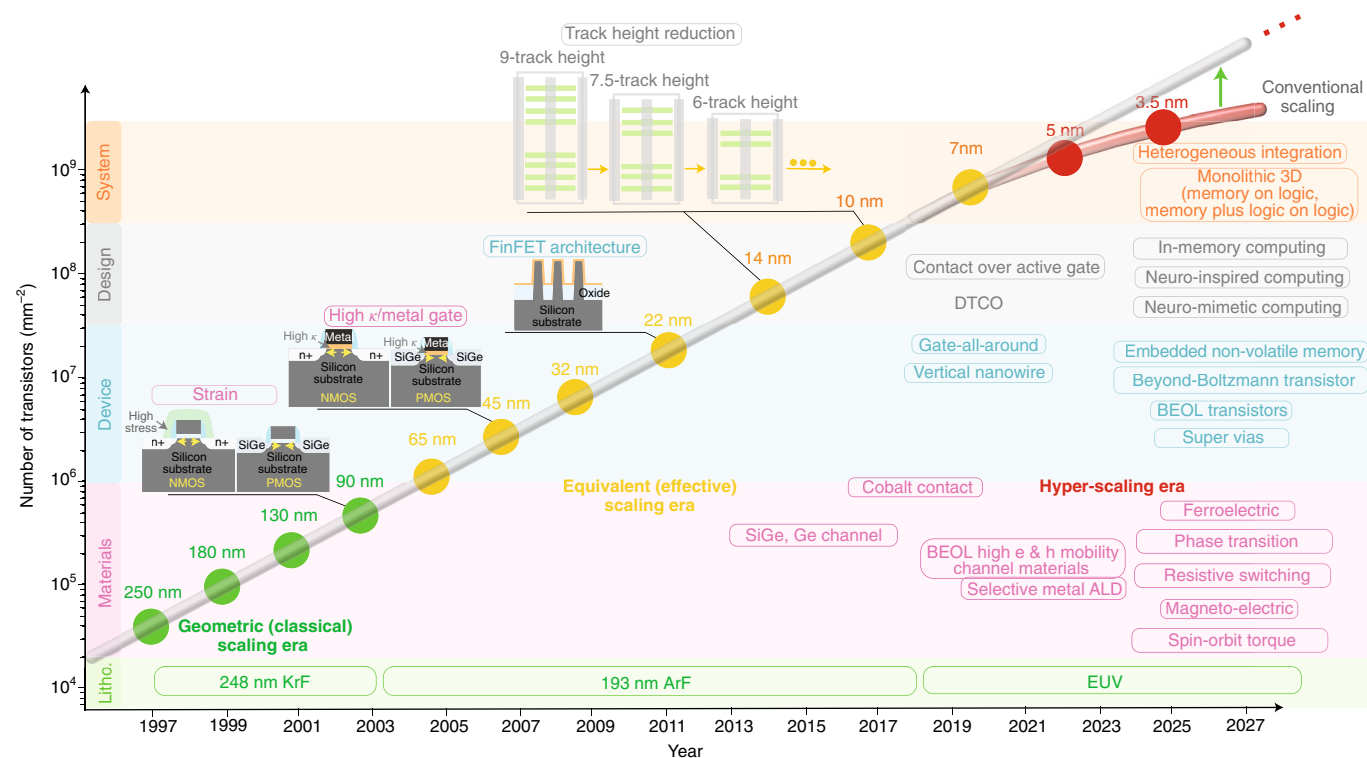


Fig. 1 | Three eras of CMOS technology scaling. Past, present and future trend of transistor density scaling, depicting three distinct eras of scaling: geometric (or classical) scaling, equivalent (or effective) scaling, and hyper-scaling (or functional diversification). Proportional scaling of the various aspects of the transistor such as gate oxide, junctions, channel doping and physical gate length characterized the era of geometric scaling. The equivalent scaling era saw the introduction of unconventional materials such as silicon-germanium, hafnium-based high- κ dielectric and non-planar device structures such as FinFETs that scaled the effective mobility, the electrical gate oxide thickness and the effective transistor width, respectively. In the future, innovations in materials, devices with both logic and memory functions and heterogeneous integration technologies will enable the era of hyper-scaling in advanced electronics. Litho., lithography; BEOL, back end of line; ALD, atomic layer deposition.

words, Boltzmann statistics dictate that, to change the current in a MOSFET by one order of magnitude, one must apply a minimum of 60 millivolts. In order to sustain the future exponential growth of semiconductor electronics, it is imperative to explore novel physical principles that can go beyond this Boltzmann limit.

Material systems with internal order could play a critical role in this regard. This is based, in particular, on the fact that a correlated material has a large degree of order and, hence, a much lower entropy. Thus, if the state of such a material could be switched from one low-entropy state to another low-entropy state, the total change in free energy (and thereby the energy dissipation) could be much lower, even below the otherwise fundamental limit of $Nk_B T \ln 2$, where N is the number of state variables participating in the switching process, k_B is the Boltzmann constant and T is the temperature¹⁰. In addition, in correlated materials such as ferroelectrics and insulator-metal phase transition materials, the state transition could occur through a subtle change in electronic configuration facilitated via collective interactions (for example, electron-electron or electron-lattice). Therefore, the switching timescale could, in principle, be very fast, thereby dramatically improving the energy-latency figure of merit.

One example of a sub-Boltzmann field-effect transistor is a negative-capacitance ferroelectric FET (FeFET; Fig. 2a). A ferroelectric material stores energy from phase transition and in doing so it lends itself to be biased at a state where its capacitance is negative^{11,12}. When such a negative capacitance is added in series to the gate of a transistor, it is possible to reduce the subthreshold swing below the thermal limit of 60 mV per decade without modifying the transport physics of the FET. Not having to change the transport physics

means that the ON current can be high while the supply voltage can be reduced simultaneously. The negative capacitance can essentially amplify the applied gate voltage electrostatically, a phenomenon that we call the amplified field effect. Experimental demonstrations have shown improved subthreshold swing and short channel effects in transistors that utilize the negative capacitance effect. For instance, negative-capacitance field-effect transistor (NCFET) operation in a commercially viable 14 nm node FinFET platform has been demonstrated, showing wafer-scale integration and functional ring oscillators operating at high speed¹³. This suggests that there is a plausible pathway for NCFETs for beyond-Boltzmann operation in a high-performance integrated circuit framework.

Complementing the concept of negative capacitance FeFet, another sub-Boltzmann FET — called phase FET (PFET) — has been proposed based on abrupt insulator-to-metal phase transitions (IMT) in correlated oxides (Fig. 2b). In IMT materials that exhibit strong correlation, such as VO_2 (ref. 14), the collective response to external perturbation (temperature, pressure and electrical stimulus) manifests in the form of ‘melting’ of carriers, causing a remarkable electronic phase transformation where the electrons localized at atomic sites change to an itinerant state. This phase transformation amplifies the free-carrier concentration and, in the case of VO_2 , manifests as a sharp rise in conductivity up to five orders in magnitude at ~ 340 K. Similar to the negative capacitance effect in FeFets, the PFET exploits the negative differential resistance induced across the correlated material as the abrupt phase transition takes place. This creates an internal carrier amplification effect that facilitates steep switching in PFET beyond the Boltzmann limit and leads to

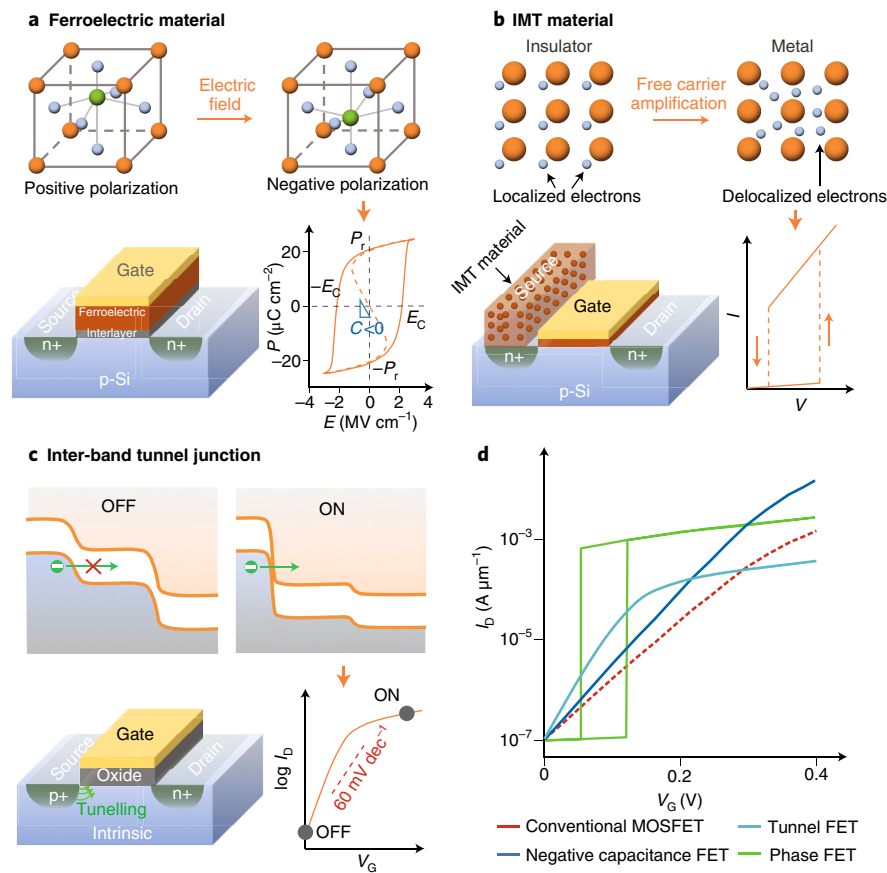


Fig. 2 | Beyond-Boltzmann transistor concepts. a–c, Schematic representations of negative capacitance (C) FET (**a**), phase transition FET (**b**) and tunnel FET (**c**). **d,** The expected current voltage transfer characteristics for each device. The negative capacitance FET takes advantage of the negative voltage drop across the ferroelectric to step up the channel potential and amplify the channel charge. The PFET exploits the abrupt IMT in IMT materials to amplify the carrier concentration in the source region as the transistor turns on. The TFET requires the replacement of the traditional p–n junction with the reverse biased tunnel junction, and harnesses the inter-band quantum-mechanical tunnelling process to filter out the high-energy tail of the carrier distribution in the source.

enhanced ON state to OFF-state current ratios over conventional field-effect transistors. Recent experimental demonstrations show that such IMT materials can be integrated with both silicon and non-silicon channel FinFETs to demonstrate low-voltage complementary n-type and p-type transistor operation with sub-thermal switching slopes, albeit with hystereses¹⁵.

In both instances of NCFET and PFET, novel materials are incorporated into the baseline FinFET architecture in order to enhance its performance and energy efficiency. Another genre of sub-Boltzmann transistors called tunnel field-effect transistors (TFETs) necessitate a more radical change to the underlying transistor structure. Figure 2c depicts how, in TFETs, the p–n junction diode at the source-channel junction of a MOSFET or FinFET is replaced by a reverse-biased tunnel junction. The gate voltage modulates the width of the tunnel junction and the states available for tunnelling within the Fermi window, and the TFET can exhibit sub- kT/q switching in its transfer characteristics at the onset of tunnelling. A decade of research on TFETs spanning a variety of material systems — such as silicon, silicon–germanium, germanium–tin, compound semiconductors and, more recently, heterostructures utilizing stacks of two-dimensional materials^{16–20} — highlight the steady progress in the improvement of the ON-state performance, the demonstration of sub- kT/q switching slope, and ultra-low OFF-state current in separate devices. However, demonstration of all three device attributes in a single device — along with complementary n-channel and p-channel operation in an integrated TFET-based integrated circuit technology — remains elusive²¹.

Figure 2d shows the expected transfer characteristics of NCFETs, PFETs and TFETs, where functional materials such as ferroelectric and phase transition oxides as well as gate controlled inter-band tunnel junctions can lead to marked improvement in the ON-to-OFF state current ratios for energy-efficient electronics, in particular at low operating voltages. It is anticipated that, although the superior sub-threshold characteristics of sub-Boltzmann FETs can help in improving the energy-latency at low voltages, the on-current at higher supply voltages (close to one volt) may still lag behind that of the conventional Boltzmann FETs. This means that, while steep slope transistors can help power up a lot more cores in a multi-core processor while running parallel workloads, we will still need conventional complementary metal–oxide semiconductor (CMOS) technology to accelerate sequential applications because the latter can sustain higher frequency at higher supply voltage. Thus, heterogeneous device technologies incorporating both traditional FETs (including beyond-silicon channel transistors) as well as sub-Boltzmann FETs will offer the best of both scenarios. However, we need careful energy allocation management schemes at the software and the micro-architectural level to maximize the overall energy-latency performance across a range of parallel and sequential workloads²².

Bringing down the memory wall

There has been an overwhelming focus by the device community in the high-performance logic space on improving the raw

performance and the energy efficiency of logic transistors, and the processor speed. On the other hand, the device community in the memory space, such as DRAM, has made density and cost the primary focus of technology development. As a result, the semiconductor industry evolved in two directions: the logic technology community focused on faster clock speed, whereas the DRAM community targeted ever increasing capacity. In the course of two decades, between 1980 and 2000, the processor speed increased by 60% every year, whereas the DRAM access time improved by 10% annually. The original memory wall problem highlighted this growing performance gap between fast processors and relatively slow memory. Since 2005, as the processor clock speed stalled due to the active power dissipation limit, the memory latency also flatlined. Over the past decade, with the increase in the number of processor cores per chip and the rise of data intensive computing workloads (such as big data analytics, massively data-parallel graphics processing, image classification and language processing using deep and recurrent neural networks)^{23,24} the memory access speed, memory bandwidth and memory energy have again become the critical bottleneck limiting system performance²⁵.

So, what technology breakthroughs do we need in the next era of hyper-scaling to enable the system to scale efficiently, bring down the memory wall and meet the demands of data intensive computing applications? The expansion of on-die embedded memory (that is, SRAM) capacity coupled with multi-threading, out-of-order instruction execution and deeper logic pipeline is one way for chip architects to hide the processor-memory performance gap. Evolutionary approaches on the hardware front include three-dimensional stacking of memory dies directly on top of processor cores that are interconnected with one another by means of through silicon vias (TSVs). Others pursue more adventurous routes of stacking multiple memory dies on top of logic and using contactless methods (for example, capacitive or inductive coupling) for high-bandwidth processor-to-memory die-to-die communication^{26,27}.

Looking ahead, we foresee two avenues that may bring down the memory wall for future data intensive applications: co-locating compute engines and dense memory blocks beyond SRAM together to enable high-bandwidth data traffic between the two; and blurring the boundary between logic and memory by designing devices with merged memory and logic functions such that computation can be embedded within the memories themselves.

SRAM²⁸ and DRAM²⁹ are the primary options for embedded memories today. Both of these memories are volatile and require power supply at all times to hold on to their information state. Although DRAM has a significant advantage in density over SRAM, it is also slower than the latter and needs periodic refresh leading to a serious power dissipation problem for embedded applications. SRAM, on the other hand, is built using six carefully sized CMOS transistors, has a much larger cell area, and consumes more standby leakage power than DRAM, thereby limiting the achievable size of the embedded memory. Reducing the refresh power in DRAM and the leakage power in SRAM remain a key goal of the device community. Beyond that the device community is earnestly looking for alternate forms of embedded non-volatile memories (eNVMs) with cell size and speed similar to DRAM but with zero stand-by power (ignoring the peripheral circuitry) that can be co-integrated with high performance logic.

Single-transistor flash memory, with a floating gate or charge-trapping layer, is a potential option for eNVM^{30,31}. However, it requires programming pulses of high amplitude (much higher than the logic compatible supply voltage) and long duration to charge up the floating gate or to fill the trap sites with electrons. Furthermore, flash memory has limited endurance properties; thus, embedded flash memory has mostly been delegated for applications with few write operations, such as for storing code for instant system boot-up from the powered down state.

More intriguing options arise from the recent advancement in emerging eNVM devices. There are five genres of emerging eNVM devices: spin-transfer torque magnetic random-access memory (STT-MRAM)^{32,33}; ferroelectric random-access memory (FeRAM) with one transistor — one capacitor configuration³⁴; single-transistor FeFET memory^{35,36}; phase-change random-access memory (PCRAM)^{37,38}; and resistive random-access memory (RRAM). Three-dimensional versions of RRAM/PCRAM can be further categorized into either vertical RRAM/PCRAM³⁹ or cross-point RRAM/PCRAM⁴⁰ depending on how they are stacked in the vertical dimension.

STT-MRAM and PCRAM are the most mature device technologies to date amongst the emerging eNVM candidates. In STT-MRAM, a spin-transfer torque is used to flip the orientation of a magnetic layer in a magnetic tunnel junction stack using a spin-polarized current. Although STT-MRAM is a significantly more scalable option than its traditional MRAM counterpart, the write current density and, hence, write power is still quite high for embedded memory applications. Novel device physics such as voltage-controlled magnetic anisotropy, spin-orbit torque (SOT) switching and magnetoelectric coupling provide interesting pathways towards reducing the write current density and improving the write speed of future MRAM devices. Phase-change memory (PCM) is based on a reversible phase transformation between the high-resistance amorphous state and the low resistance crystalline state of a chalcogenide glass, which is triggered by Joule heating and cooling of the material using current pulses. PCRAM memory exhibits a higher resistance ratio than that of STT-MRAM and thus has a higher read margin. Nevertheless, they have a higher write energy and longer write latency than STT-MRAM.

RRAM is similar to PCRAM and expected to overcome the short-comings of PCRAM in terms of write energy and write latency. RRAM involves a dielectric that is normally in the insulating state but can be made to electrically conduct via an atomically thin filament of either oxygen vacancies (oxide RRAM) or metal cations (conducting bridge RRAM) on application of a voltage. The voltages required for the one-time electroforming process in RRAM can be very high; however, the programming voltage in RRAM is compatible with the logic supply voltage, making it an attractive option for eNVM. But, RRAM suffers from cycle-to-cycle and device-to-device variation due to the stochastic nature of the conducting filament formation process. Current research is focused on how to form filaments in a deterministic fashion to make RRAM a viable eNVM option. Three-dimensional integration of RRAM either as vertical RRAM (similar to 3D NAND flash) or as cross-point array can enable dense on-chip embedded memory of sufficient size to mitigate the memory wall problem. A key challenge for 3D RRAM remains in the integration of a two-terminal selector or access device with sufficient non-linearity in its switching characteristics to suppress sneak path leakage currents, particularly in large arrays⁴¹.

FeRAM is another non-volatile memory candidate that combines the fast read and write access of DRAM cells, consisting of a ferroelectric capacitor and a transistor. FeRAM mostly uses lead zirconate titanate (also known as PZT) as the ferroelectric material, the thickness of which is hard to scale leading to a high programming voltage and longer latency than DRAM. Recent discovery of ferroelectricity in ultra-thin layers of doped hafnium dioxide (doped HfO₂), which is used extensively today in the gate stack of leading edge CMOS logic transistors as high- κ dielectric, has markedly changed the ferroelectric memory landscape. Single-transistor-based FeFET memory has been demonstrated and integrated with a leading edge CMOS process³⁵. Doped hafnia retains the ferroelectric property when its thickness is scaled to less than five nanometres and exhibits an order-of-magnitude-higher coercive field than its perovskite counterpart. Both of these factors reduce the adverse

	eSRAM	eDRAM	eFLASH	STT-MRAM	FeRAM	FeFET	PCRAM	RRAM	Vertical RRAM	Crossbar RRAM
Cell size	120–150 F^2	10–30 F^2	10–30 F^2	10–30 F^2	10–30 F^2	10–30 F^2	10–30 F^2	10–30 F^2	4 F^2/N	4 F^2/N
Cell structure	6T	1T–1C	1T	1T–1MTJ	1T–1C	1T	1T–1PCM	1T–1R	1S–1R	1S–1R
Non-volatility	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Write voltage	<1 V	<1 V	~10 V	<1.5 V	<3 V	<4 V	<3 V	<3 V	<4 V	<3 V
Write energy	~fJ	~10 fJ	~100 pJ	~1 pJ	~0.1 pJ	~0.1 pJ	~10 pJ	~1 pJ	~10 pJ	~1 pJ
Standby power	High	Medium	Low	Low	Low	Low	Low	Low	Low	Low
Write speed	~1 ns	~10 ns	0.1–1 ms	~5 ns	~10 ns	~10 ns	~10 ns	~10 ns	~100 ns	~50 ns
Read speed	~1 ns	~3 ns	~10 ns	~5 ns	~10 ns	~10 ns	~10 ns	~10 ns	~1 μ s	~50 ns
Endurance	10^{16}	10^{16}	10^4 – 10^6	10^{15}	10^{14}	$>10^5$	$>10^{12}$	$>10^7$	$>10^7$	$>10^8$

Fig. 3 | Embedded volatile and non-volatile memory. Key device parameters and performance metrics comparing various embedded memory candidates^{28–41}. FG, floating gate; FE, ferroelectric; GST, GeSbTe; F , feature size; N , number of stacked layers.

impact of the depolarization field arising from the underlying channel capacitance on the retention characteristics of the FeFETs. Figure 3 summarizes the key device parameters of all the embedded memory candidates.

It is fair to say that the optimum eNVM is yet to be determined, but its importance for future data-abundant computing cannot be stressed enough. A suitable high-density eNVM with logic compatible programming and erase voltages, fast enough write and read time and high enough endurance will be a critical resource for hyper-scaling, complementing today's SRAM-based memory.

In many cases, it is possible to take a further step in trying to blur the boundary between the logic and the memory, and embed computation within the memory itself. For example, the inherent parallelism of the cross-point memory architecture (applicable for PCRAM, RRAM and FeFET) lends itself to efficient matrix–vector multiplication — a basic compute function for learning deep representations across a variety of datasets. A subset of these eNVM devices (for example, RRAM and FeFET) display multiple stable conductance states and can be exploited to implement analogue weight cells supporting high-precision weight stores and updates, which can significantly accelerate the online learning rate at the hardware level. Learning is the most computationally heavy task in today's at-scale deep neural networks — with a multitude of hidden layers — and takes a lot more computational resource than that required for inference.

Future research will also focus on how to exploit the stochastic aspect of some of the emerging non-volatile memory devices such as STT-MRAM, RRAM and FeFETs. The idea is to extend their applications beyond traditional machine learning. Here, the goal is to emulate the biologically plausible neurosynaptic dynamics specifically targeting unsupervised learning. Research should lead to a more in-depth understanding of how to exploit their switching dynamics, particularly the stochastic nature of state transition, to implement neuromorphic computational primitives, such as long-term and short-term plasticity, stochastic weight updates, and leaky integrate and fire neurons, with the ultimate goal of deploying large-scale neuro-inspired and neuromimetic compute engines.

Exploring the third dimension

The ability of hardware to scale efficiently and effectively through the addition of resources to meet the demand of future computing

workloads involves not only adding memory-containing layers close to or on top of logic, but adding extra layers of logic on top of logic²⁵. Broadly speaking, there are two process integration approaches, parallel and sequential, to add additional layers of logic. The TSV technology is an example of the parallel integration scheme. In this scheme, the stacked wafer is processed separately and subsequently mounted on top of another processed wafer. The logic blocks reside in parallel to each other on the top and the bottom wafers, which are then electrically coupled using high-aspect-ratio TSVs. The TSV size, pitch and alignment tolerance restricts the 3D parallel integration granularity to the level of logic blocks (each block containing a few thousand transistors). On the other hand, in the 3D sequential scheme, the transistors are processed in situ sequentially on top of the bottom transistors. Needless to say, the 3D sequential integration scheme offers much flexibility to fully explore and exploit the potential of the third dimension by connecting two stacked layers at the granularity of a single transistor scale. Sequential monolithic 3D integrated circuits provide inter-layer via density that is about two to three orders of magnitude more dense than the TSV density. This results in shorter wires and mitigates the wire-related communication energy and latency problems.

There are several fundamental barriers that need to be overcome to make 3D sequential CMOS a reality for next-generation microsystems. Although devices at the bottom layer are fabricated with traditional fabrication processes, the fabrication of transistors in the upper layers are challenging since the electrical quality of the devices fabricated in the upper layers under a reduced thermal budget need to be ensured (less than 450 °C), whereas the electrical performance of the high-performance transistors in the bottom layer cannot be compromised.

There are two approaches to realizing sequential monolithic 3D integrated circuits. In the first approach, layer transfer technology — similar to the 'smart cut' process used to produce silicon-on-insulator wafers — is used to stack thin silicon device layers. Here, the inter-layer vias need to pass through the active layer, and, hence, can be much smaller than conventional TSVs, leading to extremely high via density. In another approach, the active layers are deposited and crystallized in situ in selective areas using back-end-of-line compatible processing steps. Compared to the layer transfer approach, the in situ selective active area growth approach is more cost effective. The ultimate cost–benefit analysis for monolithic 3D

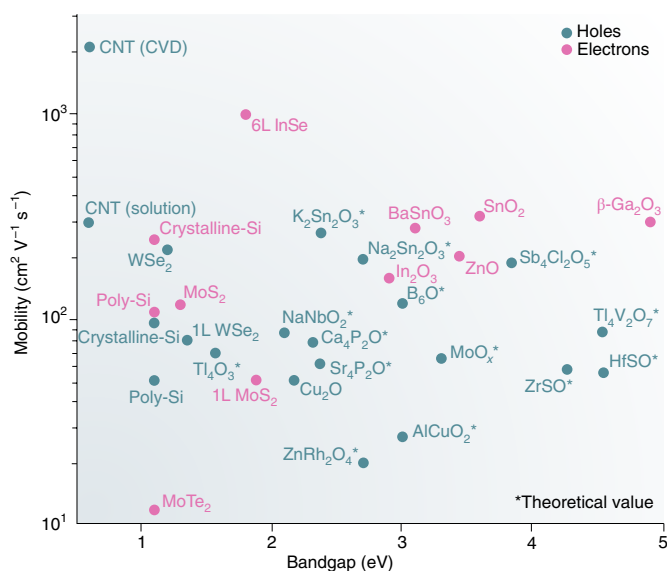


Fig. 4 | Back-end-of-line transistor options. Materials landscape for single crystal and polycrystalline silicon⁴⁴, carbon nanotubes⁴³, transition metal dichalcogenides and oxides⁴² for back-end-of-line compatible n-channel and p-channel transistor applications. High mobility ($>150 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and lower bandgap ($<1.5 \text{ eV}$) materials are suitable for logic and analogue applications; moderate mobility (50 to $150 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and intermediate bandgap (1.5 eV to 3 eV) materials are of interest for embedded memory applications; whereas ultra-wide-gap ($>3 \text{ eV}$) materials are suitable for power-delivery and -management applications. CVD, chemical vapour deposition; CNT, carbon nanotubes.

integrated circuits needs to take into account various factors such as the cost of design, verification and testing, the impact on the overall yield, and the net power-performance-area improvement.

The choice of the top-layer channel materials and their respective synthesis routes range from polycrystalline material deposition and recrystallization using ultra-fast anneal to layer transfer based on wafer-wafer bonding. Each option has its pros and cons, and significant opportunity remains for materials and device researchers to jointly innovate on novel synthesis technique(s) (such as plasma-assisted and electron-enhanced atomic layer deposition) and select the optimal set of materials to serve as the channel for the upper layer transistors.

Figure 4 shows the materials landscape for various choices as channels for the top layer transistors. The carrier mobilities span over three decades of magnitude with chemical vapour deposition (CVD) grown carbon nanotubes (CNT) exhibiting the highest values to date. The energy bandgaps may range from 0.6 eV (in the case of a single-walled CNT with a 1 nm diameter) to as high as 4.5 eV (for gallium oxide). This means that, transistors fabricated with CNTs or gallium oxide will have off-state current specifications spanning across several tens of orders of magnitude. Thus, the selection of the channel material will depend on the intended function of the upper layer transistors. For conventional high-performance logic, CNTs and crystalline silicon make sensible and practical choices. On the other hand, for an access transistor in a 3D DRAM cell, one may opt for high-mobility transition metal oxides (TMO) and transition metal dichalcogenide (TMD) channels with energy bandgaps exceeding 1.5 eV . These channel materials have the potential to reach off-state leakage currents of less than an attoampere. TMOs and TMDs with s -orbital conduction band states and wide bandgaps ($>3 \text{ eV}$) appear promising for n-channel transistors in the upper layers and sufficient for applications as access and

pass transistors. A complementary transistor solution, on the other hand, would provide more flexibility to design and implement truly monolithic 3D circuits.

High-mobility p-type TMO material remains elusive since the valence band states are derived from the oxygen p -orbitals. It is possible to 'design' extended orbital electronic states derived from the metal s -orbital above the valence band minimum formed by the oxygen p -orbital while ensuring the thermodynamic phase stability of the resulting compound. The extended hybridized electronic states derived from the transition metals, which are in the reduced valence state, have a very low hole effective mass in comparison to the large effective hole mass observed with the flat p -orbital, thereby providing high hole-mobility channel material options for p-type TMO FETs⁴².

Recent demonstrations of early prototypes of sequential monolithic 3D integrated circuits prove their practical feasibility. For example, a four-tiered 3D integrated circuit with two layers of carbon nanotube transistors and RRAM devices and two intermediate layers of dense interconnects on top of silicon CMOS logic has been demonstrated⁴³. The inter-layer interconnect density in this 3D prototype integrated circuit is three-orders-of-magnitude higher than that of TSVs found in parallel stacked 3D integrated circuits. Alternatively, a low-temperature (200°C) molecular bonding process has been employed to achieve a transfer and attachment of a monocrystalline silicon active layer to the inter-layer dielectric on a fully processed state-of-the-art silicon-on-insulator (SOI) wafer⁴⁴. Here, the top layer MOSFETs were processed at low temperature ($\leq 600^\circ \text{C}$) using a novel solid-phase epitaxy technique to activate the source drain dopants, and match the performance of the bottom layer transistors pre-fabricated at high temperature.

Novel fabrication techniques, such as selective ALD of metals and metal barrier layers, need to be developed in parallel that will address the future challenge of filling high aspect ratio inter-layer vias with ultra-low resistivity conductors in 3D integrated circuits. Last but not least, the thermal dissipation bottleneck in monolithic 3D CMOS needs to be addressed. A large number of stacked active layers with co-located logic and memory will mitigate the memory wall problem, but this will inevitably lead to increased computational power density and to elevated chip temperatures.

Current approaches toward addressing localized elevated temperatures (called hot spots) in 2D silicon chips involve attaching the active die to substrates (called heat sinks) that remove the heat either by conduction or by convection. However, they cannot be simply applied to the complex, dense 3D monolithic structures. Stacked transistors residing in the upper layers of a 3D integrated circuit suffer from severe self-heating due to the much longer thermal path imposed by the additional layers underneath. Complications arise due to the inter-layer dielectric layers and interfaces created during the sequential fabrication of the stacked transistors and the complex conduction paths that lie between the power-dense regions and the far-away heat sinks at the substrate level. It is imperative to incorporate enhanced and efficient heat spreading capabilities at the local single transistor level in addition to the die and packaging level.

In this regard, emerging two-dimensional layered materials are interesting options either as active channel layers of stacked transistors with enhanced heat dissipation capability or as local heat spreading layers in close proximity to the channel layer. The efficacy of heat management is ultimately tied to the material's thermal conductivity. Suspended single layer graphene sheets have shown record in-plane thermal conductivities of $2,000$ – $4,500 \text{ W m}^{-1} \text{ K}^{-1}$ (refs ^{45,46}). Nevertheless, despite high thermal conductivity, the small cross-section of single and few-layer graphene sheets limit the thermal conductance of these materials.

Furthermore, for local transistor level fine-grained thermal management, materials with phonon mean free paths of the same scale as the transistor dimensions are preferable such that the thermal

transport properties do not deviate significantly from their bulk values. Hexagonal boron nitride (h-BN) exhibits a phonon mean free path shorter than that in graphene, with in-plane thermal conductivity of an eleven-layer h-BN reaching $380 \text{ W m}^{-1} \text{ K}^{-1}$ near 300 K. Interestingly, the in-plane thermal conductivity in h-BN layers are two-orders-of-magnitude higher than their out-of-plane value ($1.5\text{--}2.5 \text{ W m}^{-1} \text{ K}^{-1}$). The highly anisotropic thermal conductivity is attributed to vastly different phonon dispersions in layered materials due to weak van der Waals bonding between the layers, and can be quite effective for fine-grained thermal management. The 2D h-BN layer close to the channel of the stacked transistor can transport the heat away laterally to a thermal pillar or chimney of dedicated metal vias and prevent local heating of the transistors directly underneath a region of high power density. The interface thermal resistance that exists between the 2D-layered material and the metal contact in the thermal pillar can be another limiting component in the thermal path from the 2D material to the heat sink. Thermal transport across the interface between the 2D-layered material to the contact metals is a critical aspect that needs to be understood in detail.

Thermal management in monolithic 3D integrated circuits can also be performed during the chip design time or the workload execution time or both. During chip design, thermally aware physical design for floorplanning, place and route, power delivery networks and thermal via insertion can mitigate some of the thermal issues. During run time, extensive power gating and dynamic voltage-frequency scaling techniques can be employed to mitigate the thermal hot spots and tune the chip energy-performance trade-off to specific workloads.

Embracing functional heterogeneity

The era of hyper-scaling will prominently feature heterogeneous integration of diverse chip technologies beyond high-performance logic and high-bandwidth memory, while at the same time require inter-chip data transfer bandwidth of several $\text{Tb s}^{-1} \text{ mm}^{-1}$ and energy per bit (EPB) of less than 0.1 picojoule. We envision the integration of separately manufactured electronic microcomponents into a complex assembly which, in the aggregate form, will provide not only enhanced functionality, but also superior energy, performance, form-factor and cost. The diverse microcomponents can range from discrete passives, to individual dies, stacked dies, and sub-systems like power converters, antennas and radios — all integrated into a single package or perhaps a silicon substrate.

In current technologies, the printed circuit boards (PCBs) allow for functional diversification, albeit at the cost of larger form-factor, lower chip-to-chip communication bandwidth and limited opportunity for fine-grain modularity. The inherent limit of board-level interconnect density is set by the differential stripline pair pitch at approximately $500 \mu\text{m}$. The stripline pair can support a data rate of 10 Gb s^{-1} per channel set by the channel dispersion limit and provide an aggregate bandwidth of $20 \text{ Gb s}^{-1} \text{ mm}^{-1}$. Optical interconnects featuring wavelength division multiplexing (WDM) support much higher bandwidth communication among chips. However, the electrical to optical signal conversion overhead and the laser energy efficiency limit the total achievable EPB to $5\text{--}10 \text{ pJ}$ per bit. The pitch of the polymer waveguides on the PCB is fabrication limited and is in the tens of micron range. A recent demonstration of optical inter-chip communication shows an achievable aggregate bandwidth of approximately $250 \text{ Gb s}^{-1} \text{ mm}^{-1}$ (ref. 47).

Recent advances in packaging using silicon interposers and copper-filled TSVs provide a path towards increasing the chip-to-chip interconnect bandwidth for high-performance data intensive computing applications⁴⁸. Dense interconnect pitch of the order of $15\text{--}20 \mu\text{m}$ on a silicon interposer allows for aggregate chip-to-chip communication bandwidth up to $500 \text{ Gb s}^{-1} \text{ mm}^{-1}$ and an EPB of $1\text{--}5 \text{ pJ}$ per bit. The silicon interposer enabled integration platform, commonly referred to as the 2.5D integration, can support functionally

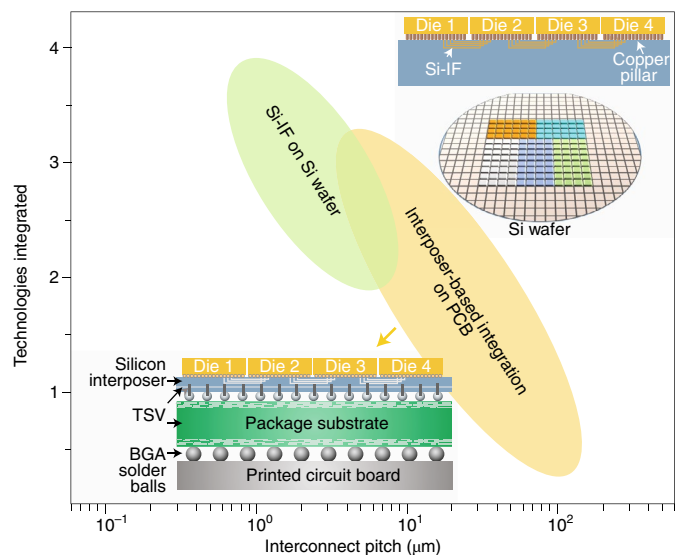


Fig. 5 | Near-monolithic performance with true heterogeneity.

Disruptive approach beyond silicon interposer-based 2.5D technology for heterogeneous integration of functionally diverse chip technologies, with a simultaneous reduction in interconnect pitch and increase in inter-die data transfer bandwidth. Si-IF is made with multiple layers of fine-pitch copper interconnect, which in turn is employed for direct assembly of functionally diverse chips onto the silicon fabric using thermocompression bonding of copper pillars. The key attributes of the Si-IF approach are fine pitch ($<5 \mu\text{m}$) inter-die interconnect, small inter-die spacing using advanced die-level pick-and-place tools, and efficient thermal dissipation capability enabled by the silicon substrate.

diverse heterogeneous technologies. It provides the system designers with the flexibility to mix and match dies that not only use different process technology nodes, but also diverse semiconductor technologies (for example, SiGe, SOI, low-voltage CMOS, high-voltage CMOS, and compound semiconductor-based heterojunction bipolar transistors). The ability to combine dies of diverse functionality onto a single package or substrate allows design houses to focus on their core design strengths, and acquire as-needed functionality (for example, high bandwidth memory, radio frequency integrated circuits, and power conversion modules) from boutique companies who can provide high-quality, proven dies. The ability to place such dies in a dense array, without the strict perimeter constraints imposed by an equivalent wire-bonded design, is attractive for system designers. However, the silicon interposer-based 2.5D integration scheme has limitations in terms of the minimum interconnect pitch that can be practically employed to connect the dies together (top right inset of Fig. 5).

Innovation in nanophotonics using silicon waveguides and their dense integration into an optical interposer, supporting $10 \mu\text{m}$ pitch closely spaced WDM channels at 10 Gb s^{-1} , has demonstrated aggregate bandwidth of almost $8 \text{ Tb s}^{-1} \text{ mm}^{-1}$. This is a very attractive proposition, but the EPB expended during the electrical to optical conversion and vice versa and the laser power consumption, still limits the adoption of optical interposers to long distance chip-to-chip communication⁴⁹.

In the era of hyper-scaling, we need an integration platform that allows integration of a multitude of diverse technologies and supports high bandwidth connectivity amongst the diverse chips, regardless of the chip-to-chip distance and expending very little energy per bit during communication. We need a radical departure from the traditional packaging roadmap. Such a revolution in heterogeneous integration may be enabled by recent advances in

new forms of heterogeneous integration fabrics, which support fine pitch, micro-aligned integration of functionally diverse dies and dielets on either rigid or flexible substrates.

Figure 5 schematically depicts the possibility of such a heterogeneous system integration scheme called the silicon interconnect fabric (Si-IF), which can implement large-scale die-to-wafer bonding, with extremely fine pitch interconnect, significantly reduced inter-die spacing compared to 2.5D packaging, and capability to integrate a multitude of diverse technologies only available with the printed circuit board today. The key idea behind the concept of Si-IF is to replace the PCB board with a silicon substrate with embedded copper-based hierarchical interconnect structure. Processor dies, memory dies and non-compute dies such as peripherals, voltage regulators, power converters, RF integrated circuits and even passive elements such as waveguides, inductors and capacitors can be bonded directly to the Si-IF.

There are several key attributes of silicon-based interconnect fabric that make it a promising technology to accelerate the next era of system integration at the hyper scale. These include: fine pitch inter-die interconnect with less than five micron pitch enabled by thermo compression bonding of copper pillars on rigid silicon wafers; small inter-die spacing enabled by state-of-the-art pick and place tools; efficient thermal dissipation capability enabled by the silicon substrate, which is a more efficient heat conductor than any PCB and package materials; and potential for integrating a multitude of diverse chip technologies (beyond processors and memories) enabled by direct mounting of bare dies onto rigid silicon substrates⁵⁰.

Disruptive innovation that supports integration of technologically and functionally diverse chips and dies on a single silicon substrate could ultimately lead to unprecedented levels of integration of heterogeneous technologies, pragmatic reuse of intellectual property blocks, and be a powerful accelerating force to usher in the era of system level hyper-scaling.

Conclusions

Semiconductor electronics is poised to enter a third era of scaling, enabled by breakthrough advances in beyond-Boltzmann transistors, novel eNVMS, sequential monolithic 3D integration and heterogeneous integration techniques. We call it the era of hyper-scaling as it requires the semiconductor technology to scale itself proportionally and appropriately to meet the energy-latency performance demand of data intensive workloads of the future.

This expectation is based on three basic premises. First, the data abundant computing applications related to machine intelligence and big data analytics have to overcome the interconnect bottleneck between logic and memory. Materials scientists and device engineers will embrace the third dimension and look for innovative ways to stack multiple layers of energy-efficient logic and high-bandwidth memory and couple them via dense and short inter-layer interconnects. This vertical CMOS technology will result in rapid acceleration of Moore's law in device packing density and will outpace the historical rate of doubling transistor count every two years. Innovations in low-temperature semiconductor materials synthesis, low-thermal-budget transistor fabrication, bottom-up fill of high-aspect-ratio vias with selective metal ALD, fine-grained thermal management solutions, and low-voltage sub-Boltzmann transistors will then follow.

Second, a subset of phase change and resistive memories, ferroelectric memories and spin-transfer torque magnetic memories will be co-integrated with CMOS. This will allow certain types of computation to be embedded within the memory block by merging logic and eNVM functions, and to stimulate the direct and efficient implementation of computation primitives like matrix-vector multiplication within the memory itself. Abundance of on-chip embedded memory with unique switching dynamics and availability of merged logic-memory fabrics will sustain the growth of the cloud computing hardware, wearable technologies and distributed sensor

networks, as well as special purpose neuro-inspired and neuromimetic application specific integrated circuits, which can accelerate on-line and real-time hardware learning and inference.

Finally, the next era of scaling will prominently feature disruptive schemes of heterogeneous integration, beyond traditional silicon interposer-based 2.5D integration approaches. This will lead to the integration of separately manufactured diverse components into a complex assembly which, in the aggregate form, will provide not only enhanced functionality, but also superior energy, latency, form-factor and cost advantages.

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Author contributions

S.S. and S.D. conceived the project, carried out the discussions and wrote the manuscripts. K.N. prepared the figures and co-wrote the section on memory benchmarking.

Competing interests

The authors declare no competing interests.

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